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HAND-HELD SURGICAL INSTRUMENTS

Abstract

A hand-held surgical instrument includes a handle housing, a shaft portion extending distally from the handle housing, and an articulation shaft. The articulation shaft may be configured to axially move within the shaft portion to articulate an end effector via a manual activation or a motor activation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/449,938 filed on Aug. 15, 2023, which is a continuation of U.S. patent application Ser. No. 17/088,769 filed on Nov. 4, 2020, now U.S. Pat. No. 11,737,747, which claims the benefit of and priority to U.S. Provisional Application No. 62/948,870 filed on Dec. 17, 2019, the entire contents of which are incorporated by reference herein.

BACKGROUND

1. Technical Field

[0002] The disclosure relates to surgical instruments. More specifically, the disclosure relates to hand-held electromechanical surgical instruments that articulate, rotate, and actuate a variety of other functions of surgical attachments, such as, for example, end effectors configured to staple tissue.

2. Background of Related Art

[0003] Electromechanical surgical instruments include a reusable handle assembly and disposable loading units and/or single use loading units, such as, for example, surgical end effectors. The end effectors are selectively connected to the handle assembly prior to use and then disconnected from the handle assembly following use in order to be disposed of or in some instances sterilized for re-use. Some surgical instruments may be capable of articulating the end effector to adjust the angle of orientation of the end effector during a surgical procedure. There are one or more drive mechanisms within the handle assembly for carrying out the articulation of the end effector and/or the operational functions of the end effector.

SUMMARY

[0004] In one aspect of the disclosure, a hand-held surgical instrument is provided and includes a handle housing, a shaft portion extending distally relative to the handle housing, a first motor disposed within the handle housing, a ball screw operably coupled to the first motor, a ball nut non-rotationally supported in the shaft portion and operably coupled to the ball screw, and a firing shaft. The firing shaft has a proximal end portion attached to the ball nut, and a distal end portion configured to fire staples from an end effector. The ball nut is configured to translate the firing shaft along a longitudinal axis defined by the shaft portion in response to a rotation of the ball screw.

[0005] In aspects, the hand-held surgical instrument may further include a battery configured to power the first motor. In aspects, the battery may be configured to power all motors, LED's, and various other electronics.

[0006] In some aspects, the handle housing may have a barrel portion, and a handle portion extending perpendicularly from the barrel portion. The battery may be supported in the handle portion. In aspects, the battery may be supported in the barrel portion.

[0007] In other aspects, the handle portion may include an upper segment fixed to the barrel portion, and a lower segment pivotably coupled to the upper segment, the battery being disposed in the lower segment.

[0008] In further aspects, the handle portion may define a plane that extends parallel with the longitudinal axis of the shaft portion. The lower segment may be configured to pivot relative to the upper segment about a pivot axis that is parallel with the plane.

[0009] In aspects, the hand-held surgical instrument may further include a printed circuit board supported in the upper segment and configured to be in electrical communication with the battery and the first motor. In aspects, the printed circuit board may be in electrical communication with

motion control circuitry.

[0010] In some aspects, the hand-held surgical instrument may further include a finger switch pivotably coupled to the upper segment. The finger switch may have an upper button and a lower button each in communication with the printed circuit board for activating the battery. In aspects, the finger switch may activate the battery and control the first motor.

[0011] In further aspects, the hand-held surgical instrument may further include a knob housing coupled to the handle housing, an articulation lever, and a first articulation shaft. The shaft portion may extend distally from the knob housing. The articulation lever may be rotationally coupled to the knob housing. The first articulation shaft may be operably coupled to the articulation lever, such that a rotation of the articulation lever translates the first articulation shaft to articulate an end effector.

[0012] In other aspects, the hand-held surgical instrument may further include a cam plate coupling the articulation lever with a proximal end portion of the first articulation shaft. The cam plate may be configured to urge the first articulation shaft in one of a proximal or distal direction upon rotation of the cam plate.

[0013] In aspects, the hand-held surgical instrument may further include an articulation locking assembly that includes a first ratchet gear operably coupled to the cam plate and a pawl engaged with the first ratchet gear. The pawl may be configured to restrict the rotation of the cam plate.

[0014] In some aspects, the first ratchet gear may be non-rotationally coupled to the articulation lever and pinned to the cam plate, such that a rotation of the articulation lever rotates the cam plate.

[0015] In further aspects, the cam plate may have a pin that extends through an elongate slot defined in the first ratchet gear. The first ratchet gear may be configured to rotate the cam plate after a delay.

[0016] In other aspects, the articulation locking assembly may include a second ratchet gear disposed between the first ratchet gear and the cam plate. The pawl may be engaged with the first and second ratchet gears.

[0017] In aspects, the first ratchet gear may have a plurality of teeth each defining an oblique surface, and the second ratchet gear may have a plurality of teeth each defining a linear surface.

[0018] In some aspects, adjacent teeth of the plurality of teeth of the first ratchet gear may define a triangular space therebetween, and adjacent teeth of the plurality of teeth of the second ratchet gear may define a rectangular space therebetween.

[0019] In further aspects, the second ratchet gear may be fixed to the cam plate, such that the cam plate and the second ratchet gear rotate simultaneously with one another.

[0020] In other aspects, the cam plate may define a first spiral slot, and the proximal end portion of the first articulation shaft may have a protuberance received in the first spiral slot.

[0021] In aspects, the hand-held surgical instrument may further include a second articulation shaft having a protuberance extending from a proximal end portion thereof. The protuberance of the second articulation shaft may be received in a second spiral slot defined in the cam plate. The first and second articulation shafts may be configured to translate in opposite directions in response to a rotation of the cam plate.

[0022] In accordance with another aspect of the disclosure, a hand-held surgical instrument is provided and includes a handle housing, a knob housing coupled to the handle housing, a shaft portion extending distally from the knob housing, an articulation lever rotationally coupled to the knob housing, a first articulation shaft, a cam plate, and an articulation locking assembly. The first articulation shaft is operably coupled to the articulation lever, such that a rotation of the articulation lever translates the first articulation shaft to articulate an end effector. The cam plate couples the articulation lever with a proximal end portion of the first articulation shaft. The cam plate is configured to urge the first articulation shaft in one of a proximal or distal direction. The articulation locking assembly includes a first ratchet gear operably coupled to the cam plate, and a pawl engaged with the first ratchet gear. The pawl may be configured to restrict the rotation of the

cam plate.

[0023] In some aspects, the cam plate may have a pin that extends through an elongate slot defined in the first ratchet gear. The first ratchet gear may be configured to rotate the cam plate after a delay.

[0024] In further aspects, the articulation locking assembly may include a second ratchet gear disposed between the first ratchet gear and the cam plate. The pawl may be engaged with the first and second ratchet gears.

[0025] In other aspects, the first ratchet gear may have a plurality of teeth each defining an oblique surface, and the second ratchet gear may have a plurality of teeth each defining a linear surface.

[0026] In aspects, adjacent teeth of the plurality of teeth of the first ratchet gear may define a triangular space therebetween, and adjacent teeth of the plurality of teeth of the second ratchet gear may define a rectangular space therebetween,

[0027] In some aspects, the second ratchet gear may be fixed to the cam plate, such that the cam plate and the second ratchet gear rotate simultaneously with one another.

[0028] In further aspects, the cam plate may define a first spiral slot, and the proximal end portion of the first articulation shaft may have a protuberance received in the first spiral slot.

[0029] In other aspects, the hand-held surgical instrument may include a second articulation shaft having a protuberance extending from a proximal end portion thereof. The protuberance of the second articulation shaft may be received in a second spiral slot defined in the cam plate. The first and second articulation shafts may be configured to translate in opposite directions in response to a rotation of the cam plate.

[0030] In accordance with yet another aspect of the disclosure, a shaft assembly for use with a handle assembly of a hand-held surgical instrument is provided. The shaft assembly includes a knob housing, a shaft portion extending distally from the knob housing, an end effector coupled to a distal end portion of the shaft portion, an articulation lever rotationally coupled to the knob housing, a first articulation shaft, and a cam plate. The first articulation shaft is operably coupled to the articulation lever, such that a rotation of the articulation lever translates the first articulation shaft to articulate the end effector. The cam plate couples the articulation lever with a proximal end portion of the first articulation shaft. The cam plate is configured to urge the first articulation shaft in one of a proximal or distal direction in response to a rotation of the cam plate.

[0031] In aspects, the shaft assembly may further include an articulation locking assembly that includes a first ratchet gear operably coupled to the cam plate and a pawl engaged with the first ratchet gear. The pawl may be configured to restrict the rotation of the cam plate.

[0032] In some aspects, the first ratchet gear may be non-rotationally coupled to the articulation lever and pinned to the cam plate, such that a rotation of the articulation lever rotates the cam plate.

[0033] In further aspects, the cam plate may have a pin that extends through an elongate slot defined in the first ratchet gear. The first ratchet gear may be configured to rotate the cam plate after a delay.

[0034] In other aspects, the articulation locking assembly may include a second ratchet gear disposed between the first ratchet gear and the cam plate. The pawl may be engaged with the first and second ratchet gears.

[0035] In aspects, the first ratchet gear may have a plurality of teeth each defining an oblique surface, and the second ratchet gear may have a plurality of teeth each defining a linear surface.

[0036] In some aspects, adjacent teeth of the plurality of teeth of the first ratchet gear may define a triangular space therebetween, and adjacent teeth of the plurality of teeth of the second ratchet gear may define a rectangular space therebetween.

[0037] In other aspects, the second ratchet gear may be fixed to the cam plate, such that the cam plate and the second ratchet gear rotate simultaneously with one another.

[0038] In accordance with yet another aspect of the disclosure, a hand-held surgical instrument is provided and includes a handle housing, a shaft portion extending distally relative to the handle

housing, a first articulation shaft supported in the shaft portion, a barrel cam, and an articulation motor. The first articulation shaft has a distal end portion configured to operably engage an end effector. The barrel cam is coupled to a proximal end portion of the first articulation shaft. The articulation motor may be operably coupled to the barrel cam and configured to rotate the barrel cam. A rotation of the barrel cam translates the first articulation shaft.

[0039] In aspects, the hand-held surgical instrument may further include a knob housing coupled to the handle housing. The knob housing may have the shaft portion extending distally therefrom. A manual rotation of the knob housing may rotate the shaft portion and the attached end effector.

[0040] In some aspects, the barrel cam may have an inner annular surface defining a helical cam slot. The proximal end portion of the first articulation shaft may have a protuberance received in the helical cam slot.

[0041] In further aspects, the hand-held surgical instrument may further include a ring gear non-rotationally coupled to the barrel cam. The articulation motor may have a motor gear operably coupled to the ring gear, such that a rotation of the motor gear results in a rotation of the barrel cam.

[0042] As used herein, the terms parallel and perpendicular are understood to include relative configurations that are substantially parallel and substantially perpendicular up to about +or -10 degrees from true parallel and true perpendicular.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0043] Embodiments of the disclosure are described herein with reference to the accompanying drawings, wherein:

[0044] FIG. 1 is a perspective view of a hand-held electromechanical surgical instrument including a handle assembly, a shaft portion coupled to the handle assembly, and a surgical end effector coupled to the shaft portion, in accordance with an embodiment of the disclosure;

[0045] FIG. 2 is an enlarged side view, with a handle housing half removed, of the surgical instrument of FIG. 1;

[0046] FIG. 3 is a rear perspective view illustrating a handle portion of the surgical instrument in an open state for removing a battery;

[0047] FIG. 4 is a side cross-sectional view of the handle portion of FIG. 3, as taken through 4-4 of FIG. 3;

[0048] FIG. 5 is a longitudinal cross-sectional view, as taken through 5-5 of FIG. 1, illustrating a drive motor and a ball screw assembly for operating a stapling function of the end effector;

[0049] FIG. 6 is a perspective view, with parts separated, of a ball nut of the ball screw assembly of FIG. 5;

[0050] FIG. 7 is a side perspective view illustrating the shaft assembly of the surgical instrument of FIG. 1;

[0051] FIG. 8 is a perspective view, with parts separated, illustrating components of an articulation assembly of the shaft assembly of FIG. 7;

[0052] FIG. 9 is an enlarged perspective view, of the indicated area of detail of FIG. 8, illustrating first and second articulation shafts of the articulation assembly of FIG. 8;

[0053] FIG. 10 is a perspective view, with parts separated, illustrating a camming mechanism of the articulation assembly of FIG. 8;

[0054] FIG. 11 is a perspective view, with parts separated, illustrating an articulation locking assembly of the shaft assembly of FIG. 7;

[0055] FIG. 12 is a top view illustrating the articulation locking assembly of FIG. 11;

[0056] FIG. 13 is a side view illustrating another embodiment of a hand-held electromechanical

surgical instrument including a handle assembly, a shaft portion coupled to the handle assembly, and a surgical end effector coupled to the shaft portion; and
[0057] FIG. **14** is a perspective view, with parts separated, illustrating components of an articulation assembly of the surgical instrument of FIG. **13**.

DETAILED DESCRIPTION

[0058] Embodiments of the presently disclosed surgical instruments are described in detail with reference to the drawings, in which like reference numerals designate identical or corresponding elements in each of the several views. As used herein the term “distal” refers to that portion of the surgical instrument, or component thereof, farther from the user, while the term “proximal” refers to that portion of the surgical instrument, or component thereof, closer to the user.

[0059] With reference to FIGS. **1** and **2**, a surgical instrument, in accordance with an embodiment of the disclosure, is generally designated as **10**, and is in the form of a powered hand-held electromechanical surgical instrument configured for selective coupling thereto of a plurality of different surgical end effectors, for example, the surgical end effector **20**. The end effector **20** is configured for actuation and manipulation by the powered hand-held electromechanical surgical instrument **10**.

[0060] The hand-held electromechanical surgical instrument **10** includes a handle assembly **12** and a shaft portion **14** extending distally from the handle assembly **12**. The shaft portion **14** is configured for selective connection with a surgical attachment, such as, for example, the end effector **20**. The handle assembly **12** has a fire switch **16** configured to actuate the various functions of the end effector **20**. In addition, the handle assembly **12** has a safety switch **18** for preventing an inadvertent actuation of the fire switch **16**. A knob housing **22** is rotationally coupled to the handle assembly **12** and configured to be manually rotated about a longitudinal axis “X” defined by the shaft portion **14** to rotate the end effector **20**. An articulation lever **24** is rotationally coupled to the knob housing **22** and is configured to articulate the end effector **20** (e.g., move the end effector **200** along a horizontal plane between a position coaxial with the shaft portion **14** and multiple positions out of alignment with the shaft portion **14**. The angular orientation of a longitudinal axis of the articulation lever relative to the longitudinal axis “X” corresponds to an angular orientation of a longitudinal axis of the end effector **20** relative to the longitudinal axis “X.” As such, the end effector **20** may articulate in the same direction and to the same angular extent as the articulation lever **24**.

[0061] With reference to FIGS. **2-4**, the handle assembly **12** includes a handle housing **26** consisting of a barrel portion **28** substantially aligned with the longitudinal axis “X,” and a handle portion **30** extending perpendicularly downward from the barrel portion **28**. The handle portion **30** includes an upper segment **32** fixed to and extending downwardly from the barrel portion **28**, and a lower segment **34** pivotably coupled to the upper segment **32**. The handle assembly **12** includes a printed circuit board **36** supported in the upper segment **32** and a battery **38** disposed in the lower segment **34**. The printed circuit board **36** is configured to be in electrical communication with the battery **38** and a drive motor **40**. The motor **40** may be wirelessly connected, connected via a wire, or otherwise electrically connected to the printed circuit board **36** and the battery **38**. The fire switch **16** may be a finger switch pivotably coupled to the upper segment **32** and has an upper button **16a** and a lower button **16b** each in communication with the printed circuit board **36** for activating the battery **38** to ultimately actuate an open/close and staple firing function of the end effector **20**.

[0062] As shown in FIGS. **3** and **4**, the upper segment **32** of the handle portion **30** has a flange **42**, such as, for example, a tab, extending downwardly therefrom, and the lower segment **34** defines a cutout **44** configured for receipt of the flange **42**. The flange **42** has an inner surface having a deflectable latch hook **46** at an end thereof, and the lower segment **34** includes a release button **48** having a tab **50** extending into the cutout **44**. The tab **50** of the release button **48** defines a notch **52** configured to selectively receive and lock with the latch hook **46** of the upper segment **32**. In some

aspects, the upper and lower segments **32**, **34** may detachably couple to one another via any suitable fastening connection, such as, for example, a bayonet-type connection. The upper and lower segments **32**, **34** are pivotably coupled to one another about a hinge **54**, such as, for example, a pivot pin. The handle portion **30** defines a plane that extends parallel with the central longitudinal axis “X” of the shaft portion **14**, such that the lower segment **34** is pivotable relative to the upper segment **32** about a pivot axis “Y” (FIG. 2) that is parallel with the plane.

[0063] With reference to FIGS. 5 and 6, the surgical instrument **10** further includes a ball screw assembly **56** operably coupled to the motor **40** for carrying out an open/close and stapling function of the end effector **20** (FIG. 1). The ball screw assembly **56** includes a ball nut **58**, a ball screw **60**, and a firing shaft **62**. The ball screw **60** is drivingly coupled to the motor **40** such that an actuation of the motor **40** results in a rotation of the ball screw **60**. The ball nut **58** is non-rotationally supported in the shaft portion **14** and operably coupled to the ball screw **60**. In particular, the ball nut **58** may have first and second planar lateral sides **58a**, **58b** that inhibit rotation of the ball nut **58** relative to the shaft portion **14**, and the ball nut **58** has an inner surface defining threading **64** that houses bearings (not shown). The bearings are captured between the threading **64** of the ball nut **58** and a threaded outer surface of the ball screw **60**.

[0064] The firing shaft **62** defines a conduit **66** through a proximal end portion **62a** thereof. The ball screw **60** extends through the conduit **66** and the proximal end portion **62a** is fixed to the ball nut **58**. In this way, the firing shaft **62** moves with the ball nut **58** as the ball nut **58** moves axially within the shaft portion **14** and relative to the ball screw **60**. The firing shaft **62** has a distal end portion **62b** configured to operably couple to an axially-driven member (not shown) of the end effector **20**. The ball nut **58** has a cap or cover **68** for containing the ball bearings therein.

[0065] FIGS. 7-10 illustrate a shaft assembly **70** of the surgical instrument **10**. The shaft assembly **70** includes the knob housing **22**, the shaft portion **14**, and the end effector **20**. The knob housing **22** supports an articulation assembly **72** configured to effect the articulation of the end effector **20** relative to the shaft portion **14**. The articulation lever **24** of the articulation assembly **72** is accessible from outside of the knob housing **22** and is configured to be manually rotated.

[0066] The articulation assembly **72** generally includes first and second articulation shafts **74**, **76** and a cam plate **82**. The first and second articulation shafts **74**, **76** are axially movable within the shaft portion **14** and each has a proximal end portion **74a**, **76a** operably coupled to the articulation lever **22**, and a distal end portion (not explicitly shown) operably coupled to opposite sides of the end effector **20**. As such, a rotation of the articulation lever **22** translates the first and second articulation shafts **74**, **76** in opposite directions to articulate the end effector **20**. The proximal end portion **74a**, **76a** of each of the articulation shafts **74**, **76** has a respective protuberance **78**, **80**. The cam plate **82** defines first and second spiral slots **82a**, **82b** for receiving the respective protuberances **78**, **80**. The spiral cam slots **82a**, **82b** are oriented so that a rotation of the cam plate **82** results in an axial movement of the first and second articulation shafts **74**, **76** in opposite directions. A helical coil **84** may be attached to the proximal end portion of the shaft portion **14** for guiding an electrical cable (not shown) thereabout that runs from the motor **40** (FIG. 5). The helical coil **84** rotates with a rotation of the shaft portion **14** and prevents the cable from bunching and eliminates the need for an electrical slip ring.

[0067] FIGS. 11 and 12 illustrate an articulation locking assembly **88** for selectively locking the articulation lever **22** in a rotational position to prevent backdriving of the articulation lever **22**. The articulation locking assembly **88** generally includes a ratchet assembly **90** operably coupling the cam plate **82** and the articulation lever **22**, and a pawl **92** engaged with the ratchet assembly **90** and configured to restrict the rotation of the cam plate **82**. The pawl **92** has a proximal end portion **92a** slidably supported on the spiral coil **84**, and a free distal end portion **92b** having an elongated distal tip **94**. The pawl **92** is slidable along a longitudinal axis defined by the pawl **92**. A detent spring **96** (FIG. 10) is provided to resiliently bias the distal tip **94** of the pawl **92** in a distal direction. In aspects, the pawl **92** may be resilient or rigid.

[0068] The ratchet assembly **90** includes a first ratchet gear **98** and a second ratchet gear **100**. The first ratchet gear **98** has a plate **102** and a stem **104** extending from the plate **102**. The stem **104** is received in a correspondingly shaped aperture (not explicitly shown) defined in the articulation lever **22** to non-rotationally couple the first ratchet gear **98** to the articulation lever **22**. The plate **102** of the first ratchet gear **98** has a plurality of teeth **106** arranged around the outer periphery of the first ratchet gear **98**. Each of the teeth **106** defines an oblique surface **108**, such that adjacent teeth **106** define a triangular space **110** therebetween configured for selective receipt of the free distal tip **94** of the pawl **92**.

[0069] The plate **102** of the first ratchet gear **98** is coupled to the cam plate **82**, such that a rotation of the articulation lever **22** rotates the cam plate **82**. For example, the cam plate **82** has a pair of pins **112**, **114** that extend through a respective elongate slot **116**, **118** defined in the first ratchet gear **98**. The elongate slots **116**, **118** define a length that is approximately 1.5 times greater than a diameter of the pins **112**, **114** of the cam plate **82**. In this way, a rotation of the first ratchet gear **98**, in response to a rotation of the articulation lever **22**, results in a rotation of the cam plate **82** after a delay.

[0070] The second ratchet gear **100** is disposed between the plate **102** of the first ratchet gear **98** and the cam plate **82**. The second ratchet gear **100** is fixed to the cam plate **82** (e.g., via the pins **112**, **114**), such that the cam plate **82** and the second ratchet gear **100** rotate simultaneously with one another. The second ratchet gear **100** has a plurality of teeth **120** arranged around an outer periphery thereof. The teeth **120** of the second ratchet gear **100** each define a linear surface **122**, such that adjacent teeth **120** of the second ratchet gear **100** define a rectangular space **124** therebetween configured for selective receipt of the distal tip **94** of the pawl **92**. The distal tip **94** of the pawl **92** may be configured to wedge into the space **124** to resist rotation of the second ratchet gear **100** relative to the pawl **92**.

[0071] The first and second ratchet gears **98**, **100** are angularly oriented relative to one another so that the triangular spaces **110** of the first ratchet gear **98** overlap with the respective rectangular spaces **124** of the second ratchet gear **100**. In aspects, the spaces **110** of the first ratchet gear **98** may assume the same shape as the spaces **124** of the second ratchet gear **100** and/or the spaces **110**, **124** may assume any suitable shape, such as, for example, arcuate.

[0072] In operation, to articulate the end effector **20**, the articulation lever **22** may be manually rotated in the direction intended for the end effector **20** to articulate. Rotation of the articulation lever **22** rotates the first ratchet gear **98**, whereby one of the oblique surfaces **108** of the teeth **106** of the first ratchet gear **98** cams the free distal tip **94** of the pawl **92** proximally and out of the space **110** between the teeth **106** of the first ratchet gear **98** and the space **124** between the teeth **120** of the second ratchet gear **100**. A tooth **106** of the first ratchet gear **98** is rotated into overlapping alignment with a space **124** defined between adjacent teeth **120** of the second ratchet gear **100**, whereby the first ratchet gear **98** engages the pins **112**, **114** of the cam plate **82** to drive a rotation of the cam plate **82**. As described, rotation of the cam plate **82** translates the first and second articulation shafts **74**, **76** in opposite directions. The opposing translation of the first and second articulation shafts **74**, **76** drives the articulation of the end effector **20**.

[0073] Due to the second ratchet gear **100** being fixed to the cam plate **82**, the second ratchet gear **100** rotates with the cam plate **82** to maintain the teeth **106** of the first ratchet gear **98** in overlapping alignment with respective spaces **124** of the second ratchet gear **100**. In this way, the distal tip **94** of the pawl **92** is maintained in a proximal position and out of the spaces **110**, **124** of the ratchet gears **98**, **100** while the articulation lever **22** is being rotated. However, upon removing the application of a rotational force on the articulation lever **22**, the resilient bias of the pawl **92** (due to detent spring **96**) will cam the first ratchet gear **98** to reposition the teeth **106** of the first ratchet gear **98** into overlapping alignment with the teeth **120** of the second ratchet gear **100**. This is caused by the first ratchet gear **98** being free to rotate relative to the second ratchet gear **100** a selected distance. Despite any backdriving force exerted on the second ratchet gear **100** via the cam

plate **82**, rotation of the second ratchet gear **100** is resisted due to the engagement of the distal tip **94** of the pawl **92** in the space **124** of the second ratchet gear **100**. More specifically, the adjacent teeth **120** of the second ratchet gear **100** capture the distal tip **94** of the pawl **92** therebetween, thereby resisting rotation of the second ratchet gear **100** and, in turn, the cam plate **82**.

[0074] FIGS. **13** and **14** illustrate another embodiment of a hand-held surgical instrument **210**, similar to the surgical instrument **10** described above. The surgical instrument **210** is different by having a powered articulation mechanism **220** rather than being manually actuated. Due to the similarities between the two surgical instruments, only those elements of the surgical instrument **210** deemed necessary to elucidate the differences from the surgical instrument **10** will be described in detail.

[0075] The surgical instrument **210** generally includes a handle housing **212**, a knob housing **222** coupled to the handle housing **212**, a shaft portion **214** extending distally from the knob housing **222**, and an end effector, such as, for example, the end effector **20**, operably coupled to a distal end portion of the shaft portion **214**. An articulation switch **216** is pivotably coupled to the handle housing **212** for actuating an articulation of the end effector **20**. The knob housing **222** may be manually rotated to thereby rotate the shaft portion **214** and the attached end effector **20** about a longitudinal axis defined by the shaft portion **214**. The shaft portion **214** has a first articulation shaft **274** and a second articulation shaft (not explicitly shown) each supported therein.

[0076] The articulation mechanism **220** is received at least partially in the knob housing **222** and includes a barrel cam **224**, a barrel cam gear **226**, and an articulation motor **228**. The barrel cam **224** consists of first and second semicircular half sections **224a**, **224b** together forming a tubular barrel cam **224**. The barrel cam **224** is received within and fixed to the barrel cam gear **226**. In some aspects, the barrel cam **224** may be monolithically formed with the barrel cam gear **226**. Each of the first and second semicircular half sections **224a**, **224b** of the barrel cam **224** defines opposing helical cam slots **230** in an inner annular surface **232** thereof. The helical cam slots **230** receive a respective protuberance extending from the proximal end portion of the first and second articulation shafts **274**. As such, a rotation of the barrel cam **224** results in axial translation of the first and second articulation shafts **274** in opposite directions.

[0077] The barrel cam gear **226** has a tubular body **226a** and a ring gear **226b** fixed about the tubular body **226a**. The articulation motor **228** has a drive shaft **236** and a drive gear **238** non-rotationally coupled to the drive shaft **236**. The drive gear **238** of the articulation motor **228** is operably coupled to the ring gear **226b** to rotate the barrel cam **224** to translate the first and second articulation shafts **274**.

[0078] In operation, an articulation switch **217** may be actuated to activate the articulation motor **228** to rotate the drive gear **238**. Rotation of the drive gear **238** drives a rotation of the barrel cam **224** via the ring gear **226b**. Due to the protuberances or pins of the articulation shafts **274** being received in the opposing helical cam slots **230** of the barrel cam **224**, rotation of the barrel cam **224** drives an axial translation of the first and second articulation shafts **274** in opposing directions to articulate the end effector **20** relative to the shaft portion **214**.

[0079] Any of the components described herein may be fabricated from either metals, plastics, resins, composites or the like taking into consideration strength, durability, wearability, weight, resistance to corrosion, ease of manufacturing, cost of manufacturing, and the like.

[0080] It will be understood that various modifications may be made to the embodiments of the presently disclosed surgical instruments including switch assemblies. Therefore, the above description should not be construed as limiting, but merely as exemplifications of embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the disclosure.

Claims

1. (canceled)
2. A hand-held surgical instrument, comprising: a handle housing; a shaft portion extending distally from the handle housing and defining a longitudinal axis, the shaft portion configured to couple to an end effector for treating tissue; an articulation mechanism operably coupled to the shaft portion for transverse articulation of the end effector relative to the longitudinal axis; and a rotatable barrel cam, comprising: a first section having a first helical cam slot oriented in a clockwise direction and configured to engage a first articulation shaft of the articulation mechanism; and a second section having a second helical cam slot oriented in a counterclockwise direction and configured to engage a second articulation shaft of the articulation mechanism; wherein rotation of the barrel cam drives opposing longitudinal translation of the first articulation shaft and the second articulation shaft for articulating motion of the end effector.
3. The hand-held surgical instrument of claim 2, further comprising a motor operably coupled to the barrel cam for driving rotation thereof.
4. The hand-held surgical instrument of claim 3, further comprising a ring gear coupled to the barrel cam, and a drive gear coupled to the motor and engaged with the ring gear, wherein driving rotation of the ring gear leads to rotation of the barrel cam.
5. The hand-held surgical instrument of claim 3, further comprising a switch disposed on the handle housing and electrically coupled to the motor for actuating an articulation of the end effector.
6. The hand-held surgical instrument of claim 2, wherein the first helical cam slot is configured to receive a protuberance of the first articulation shaft for driving longitudinal translation of the first articulation shaft in a first direction.
7. The hand-held surgical instrument of claim 6, wherein the second helical cam slot is configured to receive a protuberance of the second articulation shaft for driving longitudinal translation of the second articulation shaft in a second direction opposite the first direction.
8. The hand-held surgical instrument of claim 2, further comprising a rotatable knob housing coupled to the handle housing.
9. The hand-held surgical instrument of claim 8, wherein the shaft portion is received at least partially within the knob housing.
10. The hand-held surgical instrument of claim 8, wherein the knob housing is manually rotatable for driving rotation of the shaft portion and the end effector about the longitudinal axis.
11. The hand-held surgical instrument of claim 8, wherein the articulation mechanism is received at least partially in the knob housing.
12. A shaft assembly for use with a handle assembly of a hand-held surgical instrument, the shaft assembly comprising: a shaft portion defining a longitudinal axis and configured to couple to an end effector for treating tissue; an articulation mechanism operably coupled to the shaft portion for transverse articulation of the end effector relative to the longitudinal axis; and a rotatable barrel cam, comprising: a first section having a first helical cam slot oriented in a clockwise direction and configured to engage a first articulation shaft of the articulation mechanism; and a second section having a second helical cam slot oriented in a counterclockwise direction and configured to engage a second articulation shaft of the articulation mechanism; wherein rotation of the barrel cam drives opposing longitudinal translation of the first articulation shaft and the second articulation shaft for articulating motion of the end effector.
13. The shaft assembly of claim 12, wherein the first section comprises a first semicircular half section with an inner surface having the first helical cam slot, and wherein the second section comprises a second semicircular half section with an inner surface having the second helical cam slot.
14. The shaft assembly of claim 12, further comprising a motor with a drive gear operably coupled to the barrel cam for rotation.
15. The shaft assembly of claim 14, wherein the barrel cam further comprises a ring gear operably

coupled to the drive gear of the motor, such that the motor driving rotation of the ring gear leads to rotation of the barrel cam.

16. The shaft assembly of claim 14, further comprising a switch electrically coupled to the motor for actuating an articulation of the end effector.

17. The shaft assembly of claim 12, wherein the first articulation shaft comprises a projection receivable by the first helical cam slot for driving longitudinal translation of the first articulation shaft.

18. The shaft assembly of claim 12, further comprising a rotatable knob housing from which the shaft portion extends.

19. The shaft assembly of claim 18, wherein the shaft portion is received at least partially within the knob housing.

20. The shaft assembly of claim 18, wherein the knob housing is manually rotatable for driving rotation of the shaft portion and the end effector about the longitudinal axis.

21. The shaft assembly of claim 18, wherein the articulation mechanism is received at least partially in the knob housing.
